

Chapter 13

Trust-Region Methods for Constrained Problems

13.1 Introduction

Trust-region methods for unconstrained optimization have been discussed in Chapter 6. In this chapter we consider trust-region methods for constrained optimization.

The essential of a trust-region method is that the trial step is within a trust-region. Unlike line search methods where line searches are carried out along a search direction, trust-region algorithms compute a trial step d_k which satisfies

$$\|d_k\| \leq \Delta_k, \quad (13.1.1)$$

where $\Delta_k > 0$ is the trust-region bound at the k -th iteration, and $\|\cdot\|$ is some norm in $\mathbb{R}^n$. The fundamental belief is that the increment to the variables should not be too large and that it seems not to be a wise idea to search along a not-so-good direction (for example when step of one is not accepted in the search direction). For unconstrained optimization, a line search type algorithm normally obtains its search direction by minimizing an approximation model (for example, a quadratic model in a quasi-Newton method). Minimizing the same approximation model with the trust-region constraint:

$$\|d\| \leq \Delta_k \quad (13.1.2)$$

would give a trial step in the trust-region. Therefore it is obvious that almost all line search algorithms for unconstrained optimization can be modified to

derive corresponding trust-region algorithms.

Unfortunately, the situation for constrained cases are not the same. First, it is easy to see that we can not transform a line search algorithm for constrained optimization into a trust-region algorithm by simply adding a trust-region constraint (13.1.2) to the subproblem of a line search algorithm. Because the subproblems of most line search algorithms for constrained optimization have linear or quadratic constraints, which are approximations to the original constraints, these linear or quadratic constraints may not be consistent with the trust-region condition. For example, if the line search algorithm we have in mind is the Wilson-Han-Powell method discussed in Section 12.2, the undesirable situation is that the linearized constraints (12.2.2)–(12.2.3) may have no solutions in the trust-region (13.1.2). To overcome this infeasibility difficulty, some special considerations have to be made. There are mainly three approaches, which lead to three different types of trust-region subproblems.

The first approach is to scale the constraint violations:

$$\theta_k c_i(x_k) + d^T \nabla c_i(x_k) = 0 \quad i = 1, 2, \dots, m_e; \quad (13.1.3)$$

$$\theta_k c_i(x_k) + d^T \nabla c_i(x_k) \geq 0 \quad i = m_e + 1, \dots, m \quad (13.1.4)$$

where $\theta_k \in (0, 1]$ is a parameter (see Byrd, Schnabel and Shultz [48] and Vardi [345]). We can see that a smaller θ_k would have smaller constraint violations for the linearized constraints (13.1.3)–(13.1.4), which makes it more likely that its feasible set has a nonempty intersection with the trust-region (13.1.2). Geometrically, the parameter θ_k moves all the feasible points of the linearized constraints (12.2.2)–(12.2.3) towards the origin with the fraction of β_k . Trial steps of the trust-region algorithms that apply null space techniques can also be viewed as solutions of (13.1.2)–(13.1.4).

The second approach is replacing all the linearized constraints by a linear squares constraint. Namely, linear constraints (12.2.2)–(12.2.3) are replaced by a single constraint:

$$\sum_{i=1}^{m_e} (c_i(x_k) + d^T \nabla c_i(x_k))^2 + \sum_{i=m_e+1}^m \left(\min(0, c_i(x_k) + d^T \nabla c_i(x_k)) \right)^2 \leq \xi_k \quad (13.1.5)$$

where $\xi_k \geq 0$ is a parameter. It can be seen that if $\xi_k = 0$, the single constraint on piece-wise linear squares is equivalent to the original linearized constraints (12.2.2)–(12.2.3). The parameter ξ_k should be chosen in such a

way that the constraint (13.1.5) has a nonempty intersection with the trust-region ball (13.1.2).

The third way to overcome the inconsistency of the linearized constraints and the trust-region constraint is replacing the linearized constraints by a penalty term in the subproblem. This approach is essentially applying a trust-region algorithm for nonsmooth optimization to the corresponding penalty function.

A giant monograph on trust-region methods was published by Conn, Gould and Toint [70].

13.2 Linear Constraints

In this section we give a trust-region algorithm for linearly constrained optimization problems. The method uses trust-region conditions to define trial steps and forces all iteration points in the feasible set. The method can be considered as a combination of the feasible point method and a trust-region technique.

Consider the linearly constrained problem

$$\min_{x \in \mathcal{R}^n} f(x) \quad (13.2.1)$$

$$\text{s. t.} \quad a_i^T x = b_i, \quad i \in E, \quad (13.2.2)$$

$$a_i^T x \geq b_i, \quad i \in I. \quad (13.2.3)$$

Assume that the current iterate point x_k at the k -th iteration is feasible. The trust-region subproblem can be defined by

$$\min_{d \in \mathcal{R}^n} g_k^T d + \frac{1}{2} d^T B_k d \triangleq \phi_k(d), \quad (13.2.4)$$

$$\text{s.t.} \quad a_i^T d = 0, \quad i \in E, \quad (13.2.5)$$

$$a_i^T (x_k + d) \geq b_i, \quad i \in I, \quad (13.2.6)$$

$$\|d\|_\infty \leq \Delta_k. \quad (13.2.7)$$

It is easy to see that (13.2.4)–(13.2.7) is a quadratic programming problem, which can be solved by methods discussed in Chapter 9. Let d_k be a solution of (13.2.4)–(13.2.7). Define the ratio of actual reduction and predicted reduction by

$$r_k = \frac{f(x_k) - f(x_k + d_k)}{\phi_k(0) - \phi_k(d_k)}. \quad (13.2.8)$$

From the definition of d_k , we can easily see that $d_k = 0$ if and only if x_k is a KKT point of the original problem (13.2.1)–(13.2.3). Because all constraints are considered in the subproblem (13.2.4)–(13.2.7), the zigzagging can not happen. The following is the statement of a trust-region algorithm, assuming that the initial point x_1 is feasible.

Algorithm 13.2.1

Step 1. Given x_1 satisfying (13.2.2)–(13.2.3); given $B_1 \in \Re^{n \times n}$, $\Delta_1 > 0$, $\varepsilon \geq 0$, $k := 1$.

*Step 2. Solve (13.2.4)–(13.2.7) giving d_k ; if $\|d_k\| \leq \varepsilon$ then stop;
Compute (13.2.8);*

$$x_{k+1} = \begin{cases} x_k + d_k, & \text{if } r_k > 0 \\ x_k, & \text{Otherwise.} \end{cases} \quad (13.2.9)$$

*Step 3. If $r_k \geq 0.25$, go to Step 4;
 $\Delta_k := \Delta_k/2$, go to Step 5.*

*Step 4. If $r_k < 0.75$ or $\|d_k\|_\infty < \Delta_k$ then go to Step 5;
 $\Delta_k := 2\Delta_k$.*

*Step 5. $\Delta_{k+1} := \Delta_k$; Generate B_{k+1} ;
 $k := k + 1$; go to Step 2.*

The matrix B_{k+1} can be updated by quasi-Newton formulae. In the convergence analyses below, we assume that $\{B_k\}$ are uniformly bounded. Namely, there exists a positive constant M such that

$$\|B_k\| \leq M \quad (13.2.10)$$

holds for all k .

Theorem 13.2.2 *Assume that $f(x)$ is continuously differentiable on the feasible set and that (13.2.10) holds. If the sequence $\{x_k\}$ generated by Algorithm 13.2.1 has accumulation points, then there exists an accumulation point which is also a KKT point of the original constrained optimization problem (13.2.1)–(13.2.3).*

Proof. If the theorem is not true, we can show that

$$\lim_{k \rightarrow \infty} \Delta_k = 0. \quad (13.2.11)$$

If the above relation does not hold, there exists a positive constant $\delta > 0$, such that

$$\Delta_k \geq \delta \quad \text{and} \quad r_k \geq 0.25 \quad (13.2.12)$$

hold for infinitely many k . Define by K_0 the set of all k such that (13.2.12) hold. Without loss of generality, we assume that

$$\lim_{\substack{k \in K_0 \\ k \rightarrow \infty}} x_k = \bar{x}. \quad (13.2.13)$$

From our assumption, $\bar{x}$ is not a KKT point of (13.2.1)–(13.2.2), thus $d = 0$ is not a solution of

$$\min \quad g(\bar{x})^T d + \frac{M}{2} \|d\|_2^2 \quad (13.2.14)$$

$$\text{s.t.} \quad a_i^T d = 0, \quad i \in E, \quad (13.2.15)$$

$$a_i^T (\bar{x} + d) \geq 0, \quad i \in I, \quad (13.2.16)$$

$$\|d\|_\infty \leq \delta/2. \quad (13.2.17)$$

Let $\bar{d}$ be a solution of (13.2.14)–(13.2.17), then

$$\eta = g(\bar{x})^T \bar{d} + \frac{1}{2} M \|\bar{d}\|_2^2 < 0. \quad (13.2.18)$$

Thus, it follows from (13.2.12), (13.2.13) and (13.2.18) that

$$\phi_k(0) - \phi_k(d_k) \geq -\frac{1}{2} \eta > 0 \quad (13.2.19)$$

holds for all sufficiently large $k \in K_0$. Using (13.2.19) and the second inequality of (13.2.12) we can see that

$$f(x_k) - f(x_{k+1}) \geq -\frac{1}{8} \eta > 0 \quad (13.2.20)$$

holds for all sufficiently large $k \in K_0$. Because $\lim_{k \rightarrow \infty} f(x_k) = f(\bar{x})$, (13.2.20) can not hold for infinitely many k . This contradiction indicates that (13.2.11) must hold if the theorem is not true.

Now we suppose that the theorem is not true. The above analyses imply that (13.2.11) holds. There exists a subsequence K_1 such that

$$r_k < 0.25, \forall k \in K_1. \quad (13.2.21)$$

Assume that

$$\lim_{\substack{k \in K_1 \\ k \rightarrow \infty}} x_k = \hat{x}. \quad (13.2.22)$$

From our assumption, $\hat{x}$ is not a KKT point. Let $\hat{d}$ be a solution of the subproblem

$$\min_{d \in \mathfrak{R}^n} \quad g(\hat{x})^T d + \frac{1}{2} M \|d\|_2^2, \quad (13.2.23)$$

$$\text{s.t.} \quad a_i^T d = 0, \quad i \in E, \quad (13.2.24)$$

$$a_i^T (\hat{x} + d) \geq b_i, \quad i \in I, \quad (13.2.25)$$

$$\|d\|_\infty \leq 1, \quad (13.2.26)$$

then we have that

$$g(\hat{x})^T \hat{d} + \frac{M}{2} \|\hat{d}\|_2^2 = \hat{\eta} < 0. \quad (13.2.27)$$

Thus, because $(\Delta_k \hat{d})$ is a feasible point of the problem

$$\min_{d \in \mathfrak{R}^n} \quad g(\hat{x})^T d + \frac{1}{2} M \|d\|_2^2, \quad (13.2.28)$$

$$\text{s.t.} \quad a_i^T d = 0, \quad i \in E, \quad (13.2.29)$$

$$a_i^T (\hat{x} + d) \geq b_i, \quad i \in I, \quad (13.2.30)$$

$$\|d\|_\infty \leq \Delta_k, \quad (13.2.31)$$

we can see that

$$g(\hat{x})^T \hat{d}_k + \frac{1}{2} M \|\hat{d}_k\|_2^2 < \Delta_k \hat{\eta}, \quad (13.2.32)$$

provided that $\Delta_k \leq 1$, where $\hat{d}_k$ is a solution of (13.2.28)–(13.2.31). It follows from (13.2.22) and (13.2.32) that

$$\phi_k(0) - \phi_k(d_k) \geq -\frac{1}{2} \hat{\eta} \Delta_k \quad (13.2.33)$$

holds for all sufficiently large $k \in K_1$. From the continuously differentiable property of $f(x)$ and the uniform boundedness of $\{B_k\}$, we have that

$$Pred_k = Ared_k + o(\|d_k\|). \quad (13.2.34)$$

It can be shown from (13.2.33) and (13.2.34) that

$$\lim_{\substack{k \in K_1 \\ k \rightarrow \infty}} r_k = 1. \quad (13.2.35)$$

This contradicts (12.2.21). Therefore the theorem is true. $\square$

Similar to our analysis of the trust-region method for unconstrained optimization, Theorem 13.2.2 is still true if the condition (13.2.10) is replaced by

$$\sum_{k=1}^{\infty} \frac{1}{1 + \max_{1 \leq i \leq k} \|B_i\|} = +\infty. \quad (13.2.36)$$

From the proof of the above theorem, we can see that it is not necessary to require the trial step d_k to be the exact solution of (13.2.4)–(13.2.7). Define the projected gradient of $f(x)$ (with respect to the feasible set X) by

$$\nabla_X f(x) = \lim_{\alpha \rightarrow 0_+} \frac{P(x - \alpha \nabla f(x)) - x}{\alpha}, \quad (13.2.37)$$

where

$$P(y) = \arg \min \{\|z - y\|, z \in X\}.$$

It is not difficult to show that x^* is a KKT point of (13.2.1)–(13.2.3) if and only if

$$\nabla_X f(x^*) = 0. \quad (13.2.38)$$

From the proof of Theorem 13.2.2, we can see that Algorithm 13.2.1 remains globally convergent provided that d_k is a feasible point of (13.2.5)–(13.2.7) and satisfies

$$\phi_k(0) - \phi_k(d_k) \geq \bar{\delta} \|\nabla_X f(x_k)\| \min \left\{ \Delta_k, \frac{\|\nabla_X f(x_k)\|}{\|B_k\|} \right\}. \quad (13.2.39)$$

As for local convergence analysis, we assume that $x_k \rightarrow x^*$ and that there are only equality constraints. We also assume that the second-order sufficient conditions hold at x^* and that the Jacobian matrix $A(x^*) = \nabla c(x^*)^T \in \Re^{n \times m}$ has full column rank. Under these conditions, it is not difficult to show that Algorithm 13.2.1 is superlinearly convergent, namely

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0 \quad (13.2.40)$$

if and only if

$$\lim_{k \rightarrow \infty} \frac{\|Z^{*T}(B_k - [\nabla^2 f(x^*) - \sum \lambda_i^* \nabla^2 c_i(x^*)])Z^* Z^{*T} d_k\|}{\|d_k\|} = 0, \quad (13.2.41)$$

where $Z^* \in \Re^{n \times (n-m)}$ is a matrix satisfying $Z^{*T} A(x^*) = 0$ and $Z^{*T} Z^* = I$.

13.3 Trust-Region Subproblems

The key part of a trust-region algorithm is the calculation of the trial step d_k , which is normally a solution of a trust-region subproblem. Therefore the crucial issue of a trust-region algorithm is the construction of the trust-region subproblem. Because one of the most successful line search type methods is the sequential quadratic programming method, it is natural to consider the combination of quadratic models and trust-region technique. The combined method is usually called the TR-SQP (Trust-Region – Sequential Quadratic Programming) method. Since a trust-region constraint has the form

$$\|d\| \leq \Delta_k, \quad (13.3.1)$$

directly amalgamating (13.3.1) and the quadratic programming subproblem (12.2.1)–(12.2.3) of the sequential quadratic programming method gives the following subproblem:

$$\min_{d \in \Re^n} g_k^T d + \frac{1}{2} d^T B_k d \triangleq \phi_k(d), \quad (13.3.2)$$

$$\text{s.t.} \quad c_i(x_k) + a_i(x_k)^T d = 0, i \in E \quad (13.3.3)$$

$$c_i(x_k) + a_i(x_k)^T d \geq 0, i \in I \quad (13.3.4)$$

$$\|d\| \leq \Delta_k. \quad (13.3.5)$$

This is not a perfect way, as the constraints (13.3.3)–(13.3.5) might have no solutions. Therefore, the subproblem (13.3.2)–(13.3.5) has to be modified in order to derive a reasonable trust-region subproblem for constrained optimization.

First, we can consider a subproblem of the following type:

$$\min_{d \in \Re^n} g_k^T d + \frac{1}{2} d^T B_k d \triangleq \phi_k(d), \quad (13.3.6)$$

$$\text{s.t.} \quad \theta_k c_i(x_k) + d^T \nabla c_i(x_k) = 0, i \in E, \quad (13.3.7)$$

$$\theta_k c_i(x_k) + d^T \nabla c_i(x_k) \geq 0, i \in I, \quad (13.3.8)$$

$$\|d\| \leq \Delta_k, \quad (13.3.9)$$

where $\theta_k \in (0, 1]$ is a parameter. Subproblem (13.3.7)–(13.3.9) usually has feasible points when θ_k is sufficiently small. Geometrically, multiplying $c_i(x_k)$ by a factor θ_k is to pull the feasible points of the linearized constraints towards the original. In other words, the role of θ_k is to shift the line corresponding to the linearized constraints to a parallel line that intersects the trust-region. This technique of introducing a parameter had already been used in line search algorithms.

If $\theta_k \neq 1$, obviously the trial step d_k obtained by solving subproblem (13.3.6)–(13.3.9) may not be a feasible point of (13.3.3) and (13.3.4). In order to force d_k to be as feasible in the sense of (13.3.3)–(13.3.4) as possible, we should choose θ_k as close to 1 as possible. On the other hand, the larger the parameter θ_k , the smaller the feasible set of (13.3.7)–(13.3.9). To allow certain freedom to the subproblem, we should not choose a too large θ_k .

The minimum-norm solution of the problem

$$\min_{d \in \mathfrak{H}^n} \|(c(x_k) + A(x_k)^T d)^{(-)}\|_2 \quad (13.3.10)$$

is called the Gauss-Newton step, which is denoted by d_k^{GN} . Here $c^{(-)}$ is defined by (10.1.5)–(10.1.6). From the definition of d_k^{GN} , (13.3.7)–(13.3.9) is feasible if and only if

$$\theta_k \|d_k^{GN}\| \leq \Delta_k. \quad (13.3.11)$$

To avoid unnecessary small θ_k , it is reasonable to require

$$\theta_k \|d_k^{GN}\| \geq \delta_1 \Delta_k \quad (13.3.12)$$

if $\theta_k < 1$, where $\delta_1 \in (0, 1)$ is a given constant. For example, we can define θ_k by the formula

$$\theta_k = \begin{cases} 1, & \text{if } 2\|d_k^{GN}\| \leq \Delta_k, \\ \frac{1}{2}\Delta_k / \|d_k^{GN}\|, & \text{otherwise.} \end{cases} \quad (13.3.13)$$

An indirect way to choose the parameter θ_k is regarding $\theta = \theta_k$ as a variable. The idea of forcing θ as large as possible is achieved by a penalty term $\sigma(\theta - 1)^2$. The subproblem can be written as

$$\min_{d \in \mathfrak{H}^n, \theta \in (0, 1]} g_k^T d + \frac{1}{2} d^T B_k d + \sigma_k (\theta - 1)^2, \quad (13.3.14)$$

$$\text{s.t.} \quad \theta c_i(x_k) + d^T \nabla c_i(x_k) = 0, \quad i \in E, \quad (13.3.15)$$

$$\theta c_i(x_k) + d^T \nabla c_i(x_k) \geq 0, \quad i \in I, \quad (13.3.16)$$

$$\|d\| \leq \Delta_k, \quad (13.3.17)$$

where $\sigma_k > 0$ is a penalty parameter.

Another method to overcome the inconsistency of (13.3.3)–(13.3.5) is replacing (13.3.3)–(13.3.4) by a single constraint, which requires the sum of the squares of all linearized constraints being bounded by a certain bound:

$$\|(c_k + A_k^T d)^{(-)}\|_2^2 \leq \xi_k, \quad (13.3.18)$$

where $c_k = c(x_k) = (c_1(x_k), \dots, c_m(x_k))^T$, $A_k = A(x_k) = \nabla c(x_k)^T$, and $\xi_k > 0$ is a parameter. Thus, the subproblem has the form

$$\min_{d \in \mathfrak{R}^n} \quad g_k^T d + \frac{1}{2} d^T B_k d, \quad (13.3.19)$$

$$\text{s.t.} \quad \|(c_k + A_k^T d)^{(-)}\|_2^2 \leq \xi_k, \quad (13.3.20)$$

$$\|d\|_2^2 \leq \Delta_k^2. \quad (13.3.21)$$

It is easy to see that ξ_k must satisfy

$$\xi_k \geq \min_{\|d\|_2 \leq \Delta_k} \|(c_k + A_k^T d)^{(-)}\|_2^2, \quad (13.3.22)$$

in order to secure the feasibility of (13.3.20)–(13.3.21). Let $\bar{d}_k$ be the negative gradient direction of the function $\|(c_k + A_k^T d)^{(-)}\|_2^2$ at $d = 0$, namely $\bar{d}_k = -A_k c_k^{(-)}$, and let $\bar{\alpha}_k > 0$ be the solution of problem

$$\min_{\substack{\alpha > 0 \\ \|\alpha \bar{d}_k\|_2 \leq \Delta_k}} \|(c_k + A_k^T \alpha \bar{d}_k)^{(-)}\|_2^2. \quad (13.3.23)$$

We call $\bar{\alpha}_k \bar{d}_k$ the Cauchy point or the Cauchy step, which is denoted by d_k^{CP} . In the method of Celis, Dennis and Tapia [52],

$$\xi_k = \|(c_k + A_k^T d_k^{CP})^{(-)}\|_2^2, \quad (13.3.24)$$

while in Powell and Yuan [278], ξ_k can be any number satisfying

$$\min_{\|d\|_2 \leq b_1 \Delta_k} \|(c_k + A_k^T d)^{(-)}\|_2^2 \leq \xi_k \leq \min_{\|d\|_2 \leq b_2 \Delta_k} \|(c_k + A_k^T d)^{(-)}\|_2^2, \quad (13.3.25)$$

where $b_1 \geq b_2$ are two positive constants in $(0, 1)$.

The third type of trust-region subproblem is based on exact penalty functions. For example, based on the exact penalty function

$$P(x, \sigma) = f(x) + \sigma \|c^{(-)}(x)\|, \quad (13.3.26)$$

we can construct trust-region subproblem

$$\min_{d \in \mathfrak{R}^n} \quad g_k^T d + \frac{1}{2} d^T B_k d + \sigma_k \| (c_k + A_k^T d)^{(-)} \|, \quad (13.3.27)$$

$$\text{s.t.} \quad \|d\| \leq \Delta_k. \quad (13.3.28)$$

For this kind of subproblems, the norm in (13.3.27) and that in (13.3.28) may not be necessarily the same. For example, if we take the l_1 -norm in (13.3.27) and l_∞ -norm in (13.3.28), we obtain the subproblem

$$\begin{aligned} \min_{d \in \mathfrak{R}^n} \quad & g_k^T d + \frac{1}{2} d^T B_k d + \sigma_k \sum_{i \in E} |c_i(x_k) + \nabla c_i(x_k)^T d| \\ & + \sigma_k \sum_{i \in I} |c_i(x_k) + \nabla c_i(x_k)^T d|^{(-)} \end{aligned} \quad (13.3.29)$$

$$\text{s.t.} \quad \|d\|_\infty \leq \Delta_k. \quad (13.3.30)$$

Essentially, a trust-region algorithm based on subproblem (13.2.27)–(13.2.28) is the same as a nonsmooth trust-region algorithm for minimizing the exact penalty function (13.3.26).

13.4 Null Space Method

Consider the equality constrained problem

$$\min_{x \in \mathfrak{R}^n} \quad f(x), \quad (13.4.1)$$

$$\text{s.t.} \quad c(x) = 0. \quad (13.4.2)$$

The trust-region subproblem (13.3.6)–(13.3.9) can be written as

$$\min_{d \in \mathfrak{R}^n} \quad g_k^T d + \frac{1}{2} d^T B_k d = \phi_k(d), \quad (13.4.3)$$

$$\text{s.t.} \quad \theta_k c_k + A_k^T d = 0, \quad (13.4.4)$$

$$\|d\|_2 \leq \Delta_k. \quad (13.4.5)$$

We assume that $c_k \in \text{Range}(A_k^T)$, it follows from (13.3.11) that θ_k should satisfy

$$\theta_k \|(A_k^T)^+ c_k\|_2 \leq \Delta_k. \quad (13.4.6)$$

Let d_k be a solution of (13.4.3)–(13.4.5). It can be seen that d_k is also a solution of the following problem

$$\min_{d \in \mathfrak{R}^n} \quad \phi_k(d), \quad (13.4.7)$$

$$\text{s.t.} \quad A_k^T(d - \hat{d}_k) = 0, \quad (13.4.8)$$

$$\|d - \hat{d}_k\|_2 \leq \bar{\Delta}_k, \quad (13.4.9)$$

where

$$\hat{d}_k = -\theta_k(A_k^T)^+ c_k, \quad (13.4.10)$$

$$\bar{\Delta}_k = \sqrt{\Delta_k^2 - \|\hat{d}_k\|_2^2}. \quad (13.4.11)$$

Notice that $\hat{d}_k = \theta_k d_k^{GN}$ where d_k^{GN} is the Gauss-Newton step discussed in the previous section. Define variable $\bar{d} = d - \hat{d}_k$ and let Z_k be a matrix whose columns are an orthonormal base of the null space of A_k^T , namely $A_k^T Z_k = 0$, $Z_k^T Z_k = I$. We can then write

$$\bar{d} = Z_k u, \quad u \in \mathfrak{R}^{n-r}, \quad (13.4.12)$$

where r is the rank of A_k . Using the above relation, we can rewrite subproblem (13.4.7)–(13.4.9) in the following equivalent form:

$$\min_{u \in \mathfrak{R}^{n-r}} \quad \bar{g}_k^T u + \frac{1}{2} u^T \bar{B}_k u, \quad (13.4.13)$$

$$\text{s.t.} \quad \|u\|_2 \leq \bar{\Delta}_k, \quad (13.4.14)$$

where $\bar{g}_k = Z_k^T(g_k + B_k \hat{d}_k)$, $\bar{B}_k = Z_k^T B_k Z_k$. This is already in the form of the trust-region subproblem for unconstrained optimization, which is discussed in Chapter 6. Techniques given there can be used to solve problem (13.4.13)–(13.4.14), giving u_k . Once u_k is computed, the trial step d_k can be obtained by using $d_k = \hat{d}_k + Z_k u_k$.

We use the L_1 exact penalty function

$$P_1(x) = f(x) + \sigma_k \|c(x)\|_1 \quad (13.4.15)$$

as the merit function to decide whether the trial step d_k should be accepted. The actual reduction of the exact penalty function is

$$\text{Ared}_k = P_1(x_k) - P_1(x_k + d_k). \quad (13.4.16)$$

We define the predicted reduction by the reduction of the approximate penalty function $\phi_k(d) + \sigma_k \|c_k + A_k^T d\|_1$, namely,

$$Pred_k = \phi_k(0) - \phi_k(d_k) + \sigma_k [\|c_k\|_1 - \|c_k + A_k^T d_k\|_1]. \quad (13.4.17)$$

Assume that $f(x)$ and $c(x)$ are twice continuously differentiable and $\|B_k\|$ is bounded, then we have that

$$Ared_k = Pred_k + O(\|d_k\|_2^2). \quad (13.4.18)$$

From the definition of $\hat{d}_k$, it follows that

$$\hat{d}_k = (A_k^T)^+ A_k^T d_k, \quad (13.4.19)$$

$$d_k - \hat{d}_k = Z_k Z_k^T d_k = (I - (A_k^T)^+ A_k^T) d_k. \quad (13.4.20)$$

The step $\hat{d}_k$ is a vector in the range space of A_k , hence it is called the range space step. While the step $d_k - \hat{d}_k$ is in the null space of A_k^T , it is called the null space step. Geometrically, it is often that the range space step is vertical and the null space step is horizontal when we sketch an illustrated diagram (for example, see Figure 13.4.1). Therefore, the range space step and the null space step are called the vertical step and the horizontal step respectively.

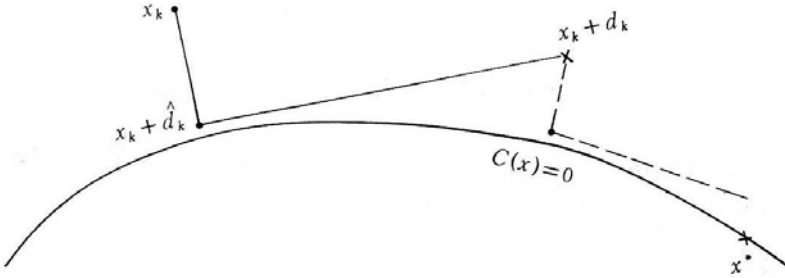


Figure 13.4.1

Using the vertical step and the horizontal step, we can decompose the predicted reduction into two parts:

$$Vpred_k = \phi_k(0) - \phi_k(\hat{d}_k) + \sigma_k (\|c_k\|_1 - \|c_k + A_k^T \hat{d}_k\|_1), \quad (13.4.21)$$

$$Hpred_k = \phi_k(\hat{d}_k) - \phi_k(d_k). \quad (13.4.22)$$

We can assume that θ_k satisfies the “not too small” condition:

$$\theta_k \|(A_k^T)^+ c_k\|_2 \geq \delta_1 \Delta_k, \text{ if } \theta_k < 1. \quad (13.4.23)$$

Assume again that we choose a sufficiently large σ_k so that

$$\sigma_k \geq \left\| A_k^+ \left(g_k + \frac{1}{2} B_k \hat{d}_k \right) \right\|_\infty + \rho. \quad (13.4.24)$$

With all these, we can show that

$$Vpred_k \geq \rho \min[\|c_k\|_1, \delta_1 \Delta_k / \|(A_k^T)^+\|_2]. \quad (13.4.25)$$

For the null space, the situation is essentially the same as that of unconstrained optimization. Applying Lemma 6.1.3, we have that

$$Hpred_k \geq \frac{1}{2} \|\bar{g}_k\|_2 \min[\bar{\Delta}_k, \|\bar{g}_k\|_2 / \|\bar{B}_k\|_2]. \quad (13.4.26)$$

Thus, we have established that there exist positive constants ρ_1, ρ_2 such that

$$\begin{aligned} Pred_k &\geq \rho_1 \min[\|c_k\|_1, \Delta_k / \|(A_k^T)^+\|_2] \\ &\quad + \rho_2 \|\bar{g}_k\|_2 \min[\bar{\Delta}_k, \|\bar{g}_k\|_2 / \|\bar{B}_k\|_2]. \end{aligned} \quad (13.4.27)$$

In practice, we can first compute $\hat{d}_k$ using the Gauss-Newton step, and then obtain u_k by solving (13.4.13)–(13.4.14) inexactly. The vector $d_k = \hat{d}_k + Z_k u_k$ satisfies the sufficient reduction condition (13.4.27).

The following is a trust-region algorithm based on null space technique.

Algorithm 13.4.1

- Step 1.* Given $x_1 \in \mathbb{R}^n$, $\Delta_1 > 0$, $\epsilon \geq 0$.
 $0 < \beta_3 < \beta_4 < 1 < \beta_1$, $0 \leq \beta_0 \leq \beta_2 < 1$,
 $\beta_2 > 0, \sigma_1 > 0, k := 1$;
- Step 2.* If $\|c_k\|_2 + \|Z_k^T g_k\|_2 \leq \epsilon$ then stop;
 If (13.4.24) is satisfied then go to Step 3. Set

$$\sigma_k = \left\| A_k^+ \left(g_k + \frac{1}{2} B_k \hat{d}_k \right) \right\| + 2\rho;$$

- Step 3.* Compute a trial step d_k satisfying (13.4.27).

Step 4. Compute $Ared_k$ and $Pred_k$ by (13.4.16)–(13.4.17);
Set $r_k = Ared_k / Pred_k$;

$$x_{k+1} = \begin{cases} x_k + d_k, & \text{if } r_k > \beta_0, \\ x_k, & \text{otherwise.} \end{cases} \quad (13.4.28)$$

Choose Δ_{k+1} such that

$$\Delta_{k+1} \in \begin{cases} (\beta_3 \|d_k\|_2, \beta_4 \Delta_k), & \text{if } r_k < \beta_2, \\ (\Delta_k, \beta_1 \Delta_k), & \text{otherwise.} \end{cases} \quad (13.4.29)$$

Step 5. Generate B_{k+1} ; Set $\sigma_{k+1} := \sigma_k$ and $k := k + 1$; go to Step 2.

In order to establish the global convergence of the above algorithm, we need the following lemma.

Lemma 13.4.2 *If d_k satisfies (13.4.27), the inequality*

$$Pred_k \geq \tau \min[\varepsilon_k, 1] \min[\Delta_k, \varepsilon_k / (1 + \|B_k\|_2)] \quad (13.4.30)$$

holds, where $\tau = \min[\rho_1 / [2 \max(1, \|A_k^+\|_2)], \rho_2 / 4]$ and

$$\varepsilon_k = \|c_k\|_2 + \|Z_k^T g_k\|_2. \quad (13.4.31)$$

Proof. If

$$\|c_k\|_2 > \frac{\Delta_k}{2\|A_k^+\|_2}, \quad (13.4.32)$$

it follows directly from (13.4.27) that

$$Pred_k \geq \rho_1 \Delta_k / 2 \|A_k^+\|_2. \quad (13.4.33)$$

Thus we see that (13.4.30) holds.

Therefore, for the rest of the proof we can assume that (13.4.32) is not true. This implies that

$$\bar{\Delta}_k = \sqrt{\Delta_k^2 - \|\hat{d}_k\|_2^2} \geq \sqrt{\Delta_k^2 - (\|A_k^+\|_2 \|c_k\|_2^2)^2} \geq \frac{1}{2} \Delta_k. \quad (13.4.34)$$

If

$$(1 + 2\|B_k\|_2 \|A_k^+\|_2) \|c_k\|_2 \leq \|Z_k^T g_k\|_2, \quad (13.4.35)$$

we can show that

$$\begin{aligned}
\|\bar{g}_k\|_2 &\geq \|Z_k^T g_k\|_2 - \|B_k\|_2 \|\hat{d}_k\|_2 \\
&\geq \|Z_k^T g_k\|_2 - \|B_k\|_2 \|A_k^+\|_2 \|c_k\|_2 \\
&\geq \frac{1}{2}(\|Z_k^T g_k\|_2 + \|c_k\|_2) = \frac{1}{2}\varepsilon_k.
\end{aligned} \tag{13.4.36}$$

This inequality, (13.4.34) and (13.4.27) indicate that (13.4.30) holds when $\tau = \rho_2/4$.

Now, we assume that inequality (13.4.35) does not hold, which implies that

$$\|c_k\|_2 \geq \frac{\varepsilon_k}{2(1 + \|B_k\|_2 \|A_k^+\|_2)}. \tag{13.4.37}$$

Consequently, we can use (13.4.27) to show that

$$\begin{aligned}
Pred_k &\geq \rho_1 \min[\varepsilon_k/[2(1 + \|B_k\|_2 \|A_k^+\|_2)], \Delta_k/\|A_k^+\|_2] \\
&\geq \frac{\rho_1}{2 \max[1, \|A_k^+\|_2]} \min[\Delta_k, \varepsilon_k/(1 + \|B_k\|_2)],
\end{aligned} \tag{13.4.38}$$

which says that (13.4.30) holds when $\tau = \rho_1/(2 \max[1, \|A_k^+\|_2])$. $\square$

The following lemma says that the norm of the trial step can not converge to zero faster than the reciprocal of the norm of quasi-Newton matrices B_k , if the iteration points are bounded away from KKT points.

Lemma 13.4.3 *Assume that $f(x)$ and $c(x)$ are twice continuously differentiable and that $\{x_k, k = 1, 2, \dots\}$ are generated by Algorithm 13.4.1. If $\|A_k\|_2$ is bounded above uniformly and*

$$\|c_k\|_2 + \|Z_k^T g_k\|_2 \geq \delta > 0 \tag{13.4.39}$$

for all k , then there exists a positive constant β_5 such that

$$\|d_k\|_2 \geq \beta_5/M_k, \quad k = 1, 2, \dots \tag{13.4.40}$$

holds for all k , where

$$M_k = \max_{1 \leq i \leq k} \|B_i\|_2 + 1. \tag{13.4.41}$$

Proof. If $\|d_k\|_2 < \Delta_k$, we can easily see that

$$\|\bar{d}_k\|_2 < \bar{\Delta}_k. \tag{13.4.42}$$

The definition of $\bar{d}_k$ and the above inequality imply that

$$\bar{g}_k + \bar{B}_k \bar{d}_k = 0. \quad (13.4.43)$$

Thus, we have

$$\|\bar{d}_k\|_2 \geq \frac{\|\bar{g}_k\|_2}{\|\bar{B}_k\|_2} \geq \frac{\|Z_k^T g_k\|_2}{\|B_k\|_2} \geq \frac{\|Z_k^T g_k\|_2}{M_k}. \quad (13.4.44)$$

From the definition of $\hat{d}_k$, we see that

$$\|\hat{d}_k\|_2 \geq \min \left[\delta_1 \Delta_k, \frac{\|c_k\|_2}{\|A_k\|_2} \right]. \quad (13.4.45)$$

Relations (13.4.44) and (13.4.45) indicate that either

$$\|d_k\|_2 \geq \delta_1 \Delta_k, \quad (13.4.46)$$

or

$$\|d_k\|_2 \geq \frac{1}{2(1 + \|A_k\|_2)} \frac{\|c_k\|_2 + \|Z_k^T g_k\|_2}{M_k}. \quad (13.4.47)$$

The boundedness of $\|A_k\|$ shows that (13.4.40) holds if (13.4.46) fails.

For the rest of the proof, we assume that (13.4.46) holds. If the lemma is not true there exists a subsequence $\{k_i\}$ such that

$$\|d_{k_i}\|_2 \geq \delta_1 \Delta_{k_i} \quad (13.4.48)$$

and

$$\lim_{i \rightarrow \infty} \Delta_{k_i} M_{k_i} = 0. \quad (13.4.49)$$

A direct consequence of the above limit is $\Delta_{k_i} \rightarrow 0$. Because M_k is monotonically increasing, we can assume that $\Delta_{k_i} < \Delta_{k_{i-1}}$ for all i . Denote $\bar{i} = k_i - 1$, (13.4.29) implies that $\|d_{\bar{i}}\|_2 \geq \Delta_{k_i}/\beta_3$, which, together with (13.4.49) and $M_{\bar{i}} \leq M_{k_i}$, shows that

$$\lim_{i \rightarrow \infty} \|d_{\bar{i}}\| M_{\bar{i}} = 0. \quad (13.4.50)$$

This shows that

$$\|d_{\bar{i}}\| \geq \delta_1 \Delta_{\bar{i}}. \quad (13.4.51)$$

This inequality guarantees the existence of a positive number $\bar{\tau}$ such that

$$Pred_{\bar{i}} \geq \bar{\tau} \|d_{\bar{i}}\|_2 \quad (13.4.52)$$

for sufficiently large i . This inequality and (13.4.18) show that

$$\lim_{i \rightarrow \infty} r_{k_i-1} = \lim_{i \rightarrow \infty} \frac{Ared_i}{Pred_i} = 1. \quad (13.4.53)$$

Therefore, for sufficient large i , we have that

$$\Delta_{k_i} \geq \Delta_{k_i-1}. \quad (13.4.54)$$

This contradicts the assumption that $\Delta_{k_i-1} < \Delta_{k_i}$. Thus, the lemma is true.

□

The following lemma is due to Powell [264], which is a very powerful tool for convergence analysis of trust-region algorithms.

Lemma 13.4.4 *Suppose that $\{\Delta_k\}$ and $\{M_k\}$ are two sequences of positive numbers. If there exist positive constants $\tau > 0$, $\beta_1 > 0$, $\beta_4 \in (0, 1)$, and a subset I of $\{1, 2, 3, \dots\}$ such that*

$$\begin{aligned} \Delta_{k+1} &\leq \beta_1 \Delta_k, & \forall k \in I; \\ \Delta_{k+1} &\leq \beta_4 \Delta_k, & \forall k \notin I; \\ \Delta_k &\geq \tau / M_k, & \forall k; \\ M_{k+1} &\geq M_k, & \forall k; \\ \sum_{k \in I} 1/M_k &< +\infty, \end{aligned} \quad (13.4.55)$$

then

$$\sum_{k=1}^{\infty} \frac{1}{M_k} < +\infty. \quad (13.4.56)$$

Proof. Let p be a positive integer satisfying

$$\beta_1 \cdot \beta_4^{p-1} < 1. \quad (13.4.57)$$

Define the set

$$I_k = I \cap \{1, 2, \dots, k\}. \quad (13.4.58)$$

Denote the number of elements of I_k by $|I_k|$. Define the set

$$J := \{k \mid k \leq p|I_k|\}. \quad (13.4.59)$$

From the monotone property of M_k and the above definition, we have that

$$\sum_{k \in J} \frac{1}{M_k} \leq p \sum_{k \in I} \frac{1}{M_k} < +\infty. \quad (13.4.60)$$

For $k \notin J$, we have that $|I_k| < k/p$, which gives $|I_{k-1}| \leq |I_k| \leq (k-1)/p$. Thus,

$$\begin{aligned} \Delta_k &\leq \beta_1^{|I_{k-1}|} \beta_4^{k-1-|I_{k-1}|} \Delta_1 \\ &\leq (\beta_1 \beta_4^{p-1})^{(k-1)/p} \Delta_1 \end{aligned} \quad (13.4.61)$$

holds for all $k \notin J$. Consequently, we have that

$$\begin{aligned} \sum_{k \notin J} \frac{1}{M_k} &\leq \sum_{k=1}^{\infty} (\beta_1 \beta_4^{p-1})^{(k-1)/p} \Delta_1 / \tau \\ &= \frac{\Delta_1}{\tau [1 - (\beta_1 \beta_4^{p-1})^{1/p}]}. \end{aligned} \quad (13.4.62)$$

Now, we can see that (13.4.56) follows from (13.4.60) and (13.4.62). $\square$

Using the above lemmas, we can prove the global convergence of Algorithm 13.4.1.

Theorem 13.4.5 *Assume that $f(x)$ and $c(x)$ are twice continuously differentiable, that all the iteration points $\{x_k\}$ generated by Algorithm 13.4.1 are in an open set S , and that $\nabla f(x)$, $\nabla^2 f(x)$, $A(x)$, $\nabla A(x)$ are bounded above on S . If $\sigma_k = \bar{\sigma}$ for all sufficiently large k , $P_1(x_k)$ is bounded below, and $\{\|A_k\|_2, \|A_k^+\|_2\}$ are uniformly bounded, and*

$$\sum_{k=1}^{\infty} \frac{1}{1 + \max_{1 \leq i \leq k} \|B_i\|_2} = \infty, \quad (13.4.63)$$

then

$$\liminf_{k \rightarrow \infty} [\|c_k\|_2 + \|Z_k^T g_k\|_2] = 0. \quad (13.4.64)$$

Furthermore, under additional assumptions that $\|B_k\|_2$ is uniformly bounded and $\beta_0 > 0$, we have that

$$\lim_{k \rightarrow \infty} [\|c_k\|_2 + \|Z_k^T g_k\|_2] = 0. \quad (13.4.65)$$

Proof. If the theorem is not true, the sequence $\{P_{\bar{\sigma}}(x_k) = f(x_k) + \bar{\sigma}\|c(x_k)\|_1\}$ is bounded below and there exists a positive constant δ such that (13.4.39) holds for all k . Define the set

$$I = \{k | r_k \geq \beta_2\}, \quad (13.4.66)$$

then it follows from Lemmas 13.4.2 and 13.4.3 that

$$\begin{aligned}
 +\infty &> \sum_{k=1}^{\infty} [P_{\bar{\sigma}}(x_k) - P_{\bar{\sigma}}(x_{k+1})] \geq \sum_{k \in I} [P_{\bar{\sigma}}(x_k) - P_{\bar{\sigma}}(x_{k+1})] \\
 &\geq \sum_{k \in I} \beta_2 \text{Pred}_k \geq \sum_{k \in I} \frac{1}{2} \beta_2 \delta \min[\Delta_k, \delta/M_k] \\
 &\geq \sum_{k \in I} \frac{1}{2} \beta_2 \delta \min[\beta_5, \delta]/M_k.
 \end{aligned} \tag{13.4.67}$$

This inequality and the previous lemma imply that

$$\sum_{k=1}^{\infty} \frac{1}{M_k} < +\infty, \tag{13.4.68}$$

which contracts (13.4.63). Therefore the theorem is true. $\square$

13.5 CDT Subproblem

Consider the subproblem (13.3.19)–(13.3.21) for the case when there are only equality constraints ($m_e = m$). It can be written as

$$\min_{d \in \mathfrak{R}^n} \quad g^T d + \frac{1}{2} d^T B d = \phi(d), \tag{13.5.1}$$

$$\text{s.t.} \quad \|A^T d + c\|_2 \leq \xi, \tag{13.5.2}$$

$$\|d\|_2 \leq \Delta, \tag{13.5.3}$$

here we omit the subscript for convenience. Such a subproblem was proposed by Celis, Dennis and Tapia [52], and is generally called the CDT subproblem.

Obviously, only when

$$\xi \geq \xi_{\min} = \min_{\|d\|_2 \leq \Delta} \|A^T d + c\|_2, \tag{13.5.4}$$

there exist feasible points for (13.5.2)–(13.5.3).

First, we consider the case when $\xi = \xi_{\min}$. It is easy to deduce from the convexity of $\|d\|_2$ that either there is only one feasible solution of (13.5.2)–(13.5.3) or that

$$\xi = \xi_{\min}^* = \min_{d \in \mathfrak{R}^n} \|A^T d + c\|_2. \tag{13.5.5}$$

The case when there is only one feasible solution of (13.5.2)–(13.5.3) requires no further consideration since this feasible point must be the solution of the CDT subproblem. This feasible point must have the following form:

$$d = -(AA^T + \lambda I)^+ c, \quad (13.5.6)$$

where $\lambda \geq 0$, and $\lambda = 0$ if $\|d\|_2 < \Delta$. $AA^T + \lambda I$ is nonsingular unless $\|d\|_2 = \Delta$. For the case when (13.5.5) holds, we have that

$$\|\hat{d}\|_2 \leq \Delta, \quad (13.5.7)$$

where $\hat{d} = -(A^T)^+ c$ is the minimal norm solution (also called the Gauss-Newton Step). Let Z be a matrix whose columns are a basis of the null space of A^T . By the variable substitution $d = \hat{d} + Zu$ as given in the previous section, problem (13.5.1)–(13.5.3) can be transformed as

$$\min_{u \in \mathbb{R}^n} \quad \bar{g}^T u + \frac{1}{2} u^T \bar{B} u, \quad (13.5.8)$$

$$\text{s.t.} \quad \|u\|_2 \leq \bar{\Delta}, \quad (13.5.9)$$

which is already in the form of the trust-region subproblem for unconstrained optimization discussed in Chapter 6.

Therefore in this section, we concentrate our attention on the case when

$$\xi > \xi_{\min}. \quad (13.5.10)$$

First we have the following necessary result.

Theorem 13.5.1 *Let d^* be a global solution of the subproblem (13.5.1)–(13.5.3). Assume that (13.5.10) holds. Then there exist nonnegative constants λ^* , μ^* such that*

$$(B + \lambda^* I + \mu^* AA^T) d^* = -(g + \mu^* Ac), \quad (13.5.11)$$

where λ^* and μ^* satisfy the complementarity conditions

$$\lambda^* [\Delta - \|d^*\|_2] = 0, \quad (13.5.12)$$

$$\mu^* [\xi - \|c + A^T d^*\|_2] = 0. \quad (13.5.13)$$

Furthermore, the matrix

$$H(\lambda^*, \mu^*) = B + \lambda^* I + \mu^* AA^T \quad (13.5.14)$$

has at most one negative eigenvalue if the multipliers λ^* and μ^* are unique.

Proof. Assumption (13.5.10) implies that the feasible region X of (13.5.2)–(13.5.3) is convex and has a nonempty interior, and we can easily prove that $LFD(d^*, X) = SFD(d^*, X)$. From the results in Chapter 8, there exist non-negative numbers λ^* and μ^* such that (13.5.11)–(13.5.13) hold. To complete the proof, we only need to prove that the matrix $H(\lambda^*, \mu^*)$ has no more than one negative eigenvalue if the multipliers λ^*, μ^* are unique.

If at most one of the constraints (13.5.2)–(13.5.3) is active at the solution d^* , the second-order sufficient condition given in Chapter 8 shows that the matrix $H(\lambda^*, \mu^*)$ has at most one negative eigenvalue.

For the rest of the proof, we assume that both constraints are active. Define the vector

$$y^* = A(c + A^T d^*). \quad (13.5.15)$$

If d^* and y^* are linearly dependent, there exists $\eta \in \Re$ such that

$$y^* = \eta d^*. \quad (13.5.16)$$

The assumption $\xi > \xi_{\min}$ implies $\eta > 0$. It follows from the uniqueness of λ^* and μ^* that $\lambda^* = \mu^* = 0$. Thus, d^* is a stationary point of $\phi(d)$. From (13.5.16) and $\eta > 0$ we see that d is a feasible direction provided that $d^T d^* < 0$. This shows that $d^T B d \geq 0$ holds for all d satisfying $d^T d^* \leq 0$, which implies that B is a semi-definite matrix.

If d^* and y^* are linearly independent, the second-order necessary condition shows that the matrix $H(\lambda^*, \mu^*)$ is positive semi-definite in the $n - 2$ dimensional subspace orthogonal to d^* and y^* . Assume that $H(\lambda^*, \mu^*)$ has two negative eigenvalues, then there exist linearly independent vectors $z_1, z_2 \in \Re^n$ such that $H(\lambda^*, \mu^*)$ is negative definite on $Span(z_1, z_2)$. The intersection of $Span(z_1, z_2)$ and the $n - 2$ dimensional subspace mentioned above is empty except the original. Therefore the matrix

$$\begin{pmatrix} z_1^T d^* & z_2^T d^* \\ z_1^T y^* & z_2^T y^* \end{pmatrix} \quad (13.5.17)$$

is nonsingular. The nonsingularity of the above matrix implies the existence of a nonzero vector $\bar{d} \in Span(z_1, z_2)$ such that

$$\|d^* + \bar{d}\|_2 = \Delta, \|c + A^T(d^* + \bar{d})\|_2 = \xi. \quad (13.5.18)$$

Relation (13.5.18) and the negative definiteness of $H(\lambda^*, \mu^*)$ on $Span(z_1, z_2)$ give that $\phi(d^* + \bar{d}) < \phi(d^*)$. This contradicts the optimality of d^* . Hence the lemma is true. $\square$

The following is a sufficient condition.

Theorem 13.5.2 *Let d^* be a feasible point of (13.5.2)–(13.5.3). If there exist $\lambda^* \geq 0$ and $\mu^* \geq 0$ such that (13.5.11)–(13.5.13) hold, and that $H(\lambda^*, \mu^*)$ is positive semi-definite, then d^* is a global solution of (13.5.1)–(13.5.13).*

Proof. Let d be any vector satisfying (13.5.2)–(13.5.3). We have that

$$\begin{aligned}
 \phi(d) &= \phi(d) + \frac{1}{2}\lambda^*\|d\|_2^2 + \frac{1}{2}\mu^*\|c + A^T d\|_2^2 \\
 &\quad - \frac{1}{2}[\lambda^*\|d\|_2^2 + \mu^*\|c + A^T d\|_2^2] \\
 &\geq \phi(d^*) + \frac{1}{2}\lambda^*\|d^*\|_2^2 + \frac{1}{2}\mu^*\|c + A^T d^*\|_2^2 \\
 &\quad - \frac{1}{2}[\lambda^*\|d\|_2^2 + \mu^*\|c + A^T d\|_2^2] \\
 &= \phi(d^*) + \frac{1}{2}\lambda^*[\Delta^2 - \|d\|_2^2] + \frac{1}{2}\mu^*[\xi^2 - \|c + A^T d\|_2^2] \\
 &\geq \phi(d^*).
 \end{aligned} \tag{13.5.19}$$

Thus, we can see that d^* is a global solution of (13.5.1)–(13.5.3). $\square$

A direct consequence of the above theorem is the following.

Corollary 13.5.3 *Assume that B is positive semi-definite. A feasible point d^* of (13.5.2)–(13.5.3) is a solution of (13.5.1)–(13.5.3) if and only if there exist $\lambda^* \geq 0$, $\mu^* \geq 0$ such that (13.5.11)–(13.5.13) hold.*

Therefore, when B is positive definite, the solution of (13.5.1)–(13.5.3) must have the form

$$d(\lambda, \mu) = -H(\lambda, \mu)^{-1}[g + \mu Ac]. \tag{13.5.20}$$

From Corollary 13.5.3 we can easily see that the following lemma holds.

Lemma 13.5.4 *Assume that B is positive definite. Then $d(\lambda, \mu)$ defined by (13.5.20) is a solution of (13.5.1)–(13.5.3) if and only if it is a feasible point of (13.5.2)–(13.5.3), and one of the following holds:*

1. $\lambda = \mu = 0$;
2. $\lambda > 0$, $\mu = 0$, $\|d(\lambda, \mu)\|_2 = \Delta$;
3. $\lambda = 0$, $\mu > 0$, $\|c + A^T d(\lambda, \mu)\|_2 = \xi$;

4. $\lambda > 0, \mu > 0$, $\|d(\lambda, \mu)\|_2 = \Delta$, $\|c + A^T d(\lambda, \mu)\|_2 = \xi$.

From the above statements, solving a convex CDT subproblem is equivalent to finding $\lambda^*, \mu^* \geq 0$ such that $d(\lambda^*, \mu^*)$ is feasible and one of the four possibilities in Lemma 13.5.4 holds.

For the case $\lambda^* = \mu^* = 0$, the solution is $d = -B^{-1}g$.

For $\mu^* = 0$ and $\lambda^* > 0$, we can solve $\bar{\psi}(\lambda, 0) = 0$ to obtain λ^* , where

$$\bar{\psi}(\lambda, \mu) = \frac{1}{\|d(\lambda, \mu)\|_2} - \frac{1}{\Delta}. \quad (13.5.21)$$

The reason for considering $\bar{\psi}(\lambda, 0) = 0$ instead of $\|d(\lambda, 0)\|_2 = \Delta$ is similar to that in Chapter 6, namely $\bar{\psi}(\lambda, 0)$ behaves more like a linear function. $\bar{\psi}(\lambda, \mu)$ as a function of λ is concave and increasing, thus we can apply Newton's iteration:

$$\lambda_+ = \lambda - \frac{\bar{\psi}(\lambda, 0)}{\bar{\psi}'_\lambda(\lambda, 0)}. \quad (13.5.22)$$

It is not difficult to show that iteration process (13.5.22) with any initial $\lambda \in [0, \lambda^*]$ will generate a monotone increasing sequence converging to λ^* .

When $\lambda^* = 0$ and $\mu^* > 0$, we define

$$\hat{\psi}(\lambda, \mu) = \frac{1}{\|c + A^T d(\lambda, \mu)\|_2} - \frac{1}{\xi}. \quad (13.5.23)$$

Similarly, we can apply Newton's method to $\hat{\psi}(0, \mu) = 0$, that is,

$$\mu_+ = \mu - \frac{\hat{\psi}(0, \mu)}{\hat{\psi}'_\mu(0, \mu)}. \quad (13.5.24)$$

When $\lambda^* > 0$ and $\mu^* > 0$, we need to solve

$$\bar{\psi}(\lambda, \mu) = 0, \quad (13.5.25)$$

$$\hat{\psi}(\lambda, \mu) = 0. \quad (13.5.26)$$

The Newton iteration for the above system is

$$\begin{pmatrix} \lambda_+ \\ \mu_+ \end{pmatrix} = \begin{pmatrix} \lambda \\ \mu \end{pmatrix} - J(\lambda, \mu)^{-1} \begin{pmatrix} \bar{\psi}(\lambda, \mu) \\ \hat{\psi}(\lambda, \mu) \end{pmatrix}, \quad (13.5.27)$$

where $J(\lambda, \mu)$ is the Jacobi matrix:

$$J(\lambda, \mu) = \begin{bmatrix} \bar{\psi}'_\lambda(\lambda, \mu) & \bar{\psi}'_\mu(\lambda, \mu) \\ \hat{\psi}'_\lambda(\lambda, \mu) & \hat{\psi}'_\mu(\lambda, \mu) \end{bmatrix}. \quad (13.5.28)$$

An algorithm based on the above analyses is given as follows.

Algorithm 13.5.5

- Step 1.* Given $g \in \mathbb{R}^n$, B positive definite, $\Delta > 0$, $\xi > \xi_{\min}$.
- Step 2.* Compute $d(0, 0)$. If $d(0, 0)$ is feasible then stop;
If $\|d(0, 0)\| \leq \Delta$ then go to Step 4;
- Step 3.* Applying (13.5.22) to solve $\bar{\psi}(\lambda, 0) = 0$ giving λ^* ;
If $d(\lambda^*, 0)$ is feasible then stop;
- Step 4.* Applying (13.5.24) to solve $\hat{\psi}(0, \mu) = 0$ giving μ^* ;
If $d(0, \mu^*)$ is feasible then stop;
- Step 5.* Applying (13.5.27) to solve (13.5.25)–(13.5.26) giving λ^*, μ^* ;
stop.

The above algorithm is in fact an enumeration of the four cases given in Lemma 13.5.4. A more direct way is to solve the system

$$\begin{pmatrix} \bar{\psi}(\lambda, \mu) \\ \hat{\psi}(\lambda, \mu) \end{pmatrix} \geq 0, \quad (\lambda, \mu)^T \begin{bmatrix} \bar{\psi}(\lambda, \mu) \\ \hat{\psi}(\lambda, \mu) \end{bmatrix} = 0 \quad (13.5.29)$$

in the nonnegative orthant $\mathbb{R}_+^2 = \{\lambda \geq 0, \mu \geq 0\}$. Such an approach to identify the Lagrange multipliers λ^* and μ^* is equivalent to solving the dual problem of (13.5.1)–(13.5.3). A truncated Newton's method based on the dual of (13.5.1)–(13.5.3) is given by Yuan [373], which is basically the Newton-Raphson method for the nonlinear system (13.5.29). The approach given by Zhang [381] is to reformulate (13.5.29) as a univariate problem. Basically it is to solve the problem

$$\hat{\psi}(\lambda(\mu), \mu) = 0 \quad (13.5.30)$$

where $\lambda(\mu)$ is defined by $\bar{\psi}(\lambda, \mu) = 0$.

13.6 Powell-Yuan Algorithm

Consider the constrained optimization problem (13.4.1)–(13.4.2). The trial step d_k is obtained by solving

$$\min_{d \in \mathbb{R}^n} \quad g_k^T d + \frac{1}{2} d^T B_k d = \phi_k(d), \quad (13.6.1)$$

$$\text{s. t.} \quad \|c_k + A_k^T d\|_2 \leq \xi_k, \quad (13.6.2)$$

$$\|d\|_2 \leq \Delta_k, \quad (13.6.3)$$

where Δ_k is the trust-region radius, ξ_k is a parameter satisfying (13.3.25). The merit function is Fletcher's differentiable function:

$$P_k(x) = f(x) - \lambda(x)^T c(x) + \sigma_k \|c(x)\|_2^2, \quad (13.6.4)$$

where $\sigma_k > 0$ is a penalty parameter, $\lambda(x)$ is the minimum norm solution of

$$\min_{\lambda \in \mathbb{R}^m} \|g(x) - A(x)\lambda\|_2. \quad (13.6.5)$$

The actual reduction is

$$Ared_k = P_k(x_k) - P_k(x_k + d_k), \quad (13.6.6)$$

and the predicted reduction is defined by

$$\begin{aligned} Pred_k = & -(g_k - A_k \lambda_k)^T d_k - \frac{1}{2} d_k^T B_k \bar{d}_k \\ & + [\lambda(x_k + d_k) - \lambda_k]^T \left(c_k + \frac{1}{2} A_k^T d_k \right) \\ & + \sigma_k (\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2), \end{aligned} \quad (13.6.7)$$

where $\bar{d}_k$ is the orthogonal projection of d_k to the null space of A_k^T , namely

$$\bar{d}_k = \bar{P}_k d_k, \quad (13.6.8)$$

$$\bar{P}_k = I - A(x_k)A(x_k)^+. \quad (13.6.9)$$

If $\|c_k\|_2 - \|c_k + A_k^T d_k\|_2 > 0$, from (13.6.7) and by increasing σ_k (if needed), we have that

$$Pred_k \geq \frac{1}{2} \sigma_k (\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2). \quad (13.6.10)$$

If $\|c_k\|_2 - \|c_k + A_k^T d_k\|_2 = 0$, d_k is the minimizer of $\phi_k(d)$ in the intersection of the trust-region and the null space of A_k^T and $Pred_k = \phi_k(0) - \phi_k(d_k)$. Thus, $Pred_k = 0$ if and only if $g_k - A_k \lambda_k = 0$.

The following algorithm is given by Powell and Yuan(1991):

Algorithm 13.6.1

Step 1. Given $x_1 \in \mathbb{R}^n$, $\Delta_1 > 0$, $\epsilon > 0$.

$0 < \tau_3 < \tau_4 < 1 < \tau_1$, $0 \leq \tau_0 \leq \tau_2 < 1$, $\tau_2 > 0$; $k := 1$.

Step 2. If $\|c_k\|_2 + \|g_k - A_k \lambda_k\|_2 \leq \epsilon$ then Stop. Otherwise solve the problem (13.6.1)-(13.6.3) which gives d_k ;

Step 3. Calculate $Pred_k$ by formula (13.6.7); If (13.6.10) is satisfied then go to Step 4; Set

$$\sigma_k := 2\sigma_k + \max \left\{ 0, \frac{-2Pred_k}{\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2} \right\}. \quad (13.6.11)$$

Recalculate $Pred_k$ by (13.6.7).

Step 4. Calculate the ratio $r_k = Ared_k/Pred_k$. Set the values

$$x_{k+1} = \begin{cases} x_k + d_k, & \text{if } r_k > 0, \\ x_k, & \text{otherwise;} \end{cases} \quad (13.6.12)$$

and

$$\Delta_{k+1} = \begin{cases} \max[4\|d_k\|_2, \Delta_k], & \text{if } r_k > 0.9, \\ \Delta_k, & 0.1 \leq r_k \leq 0.9, \\ \min[\Delta_k/4, \|d_k\|_2/2], & r_k < 0.1. \end{cases} \quad (13.6.13)$$

Step 5. Generate B_{k+1} . Set $\sigma_{k+1} := \sigma_k$. Set $k := k + 1$ and go to Step 2.

In order to establish the convergence results of the above algorithm, we make the following assumptions.

Assumption 13.6.2

1. *There exists a bounded convex closed set $\Omega \in \mathbb{R}^n$ such that $\{x_k\}$, $\{x_k + d_k\}$ are all in Ω for all k .*
2. *$A(x)$ has full column rank for all $x \in \Omega$.*
3. *The matrices $\{B_k | k = 1, 2, \dots\}$ are uniformly bounded.*

The following two lemmas provide a lower bound on the predicted reduction $Pred_k$.

Lemma 13.6.3 *The inequality*

$$\|c_k\|_2 - \|c_k + A_k^T d_k\|_2 \geq \min \left[\|c_k\|_2, \frac{b_2 \Delta_k}{\|A_k^+\|_2} \right] \quad (13.6.14)$$

holds for all k , where b_2 is introduced in (13.3.25).

Proof. If $b_2\Delta_k \geq \|(A_k^T)^+c_k\|_2$, we have $\xi_k = 0$. Thus,

$$\|c_k\|_2 - \|c_k + A_k^T d_k\|_2 = \|c_k\|_2, \quad (13.6.15)$$

which implies (13.6.14).

In the case when $b_2\Delta_k < \|(A_k^T)^+c_k\|_2$, it follows from (13.3.25) and constraint condition (13.6.2) that

$$\begin{aligned} \|c_k\|_2 - \|c_k + A_k^T d_k\| &\geq \|c_k\|_2 - \xi_k \\ &\geq \|c_k\|_2 - \left\| c_k - A_k^T \left[\frac{b_2\Delta_k}{\|(A_k^T)^+c_k\|_2} \right] (A_k^T)^+c_k \right\|_2 \\ &= \|c_k\|_2 \frac{b_2\Delta_k}{\|(A_k^T)^+c_k\|_2} \geq \frac{b_2\Delta_k}{\|A_k^+\|_2}. \end{aligned} \quad (13.6.16)$$

Thus the lemma is true. $\square$

Lemma 13.6.4 *There exists a positive constant δ_1 such that the inequality*

$$\begin{aligned} Pred_k &- \frac{1}{2}\sigma_k(\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2) + \delta_1\|d_k\|_2\|c_k\|_2 \\ &\geq \frac{1}{4}\|\bar{P}_k\bar{g}_k\|_2 \min \left[\bar{\Delta}_k, \frac{\|\bar{P}_k\bar{g}_k\|_2}{2\|B_k\|_2} \right] \\ &\quad + \frac{1}{2}\sigma_k\|c_k\|_2 \min \left[\|c_k\|_2, \frac{b_2\Delta_k}{\|A_k^+\|_2} \right] \end{aligned} \quad (13.6.17)$$

holds for all k , where we use the notation

$$\bar{g}_k = g_k + B_k\hat{d}_k, \quad (13.6.18)$$

$$\hat{d}_k = d_k - \bar{P}_k d_k = d_k - \bar{d}_k, \quad (13.6.19)$$

$$\bar{\Delta}_k = \sqrt{\Delta_k^2 - \|\hat{d}_k\|_2^2}. \quad (13.6.20)$$

Proof. The definition of $\hat{d}_k$ and $\|c_k + A_k^T d_k\|_2 \leq \|c_k\|_2$ imply the bound

$$\begin{aligned} \|\hat{d}_k\|_2 &= \|A_k A_k^+ d_k\|_2 = \|(A_k^+)^T[(c_k + A_k^T d_k) - c_k]\|_2 \\ &\leq 2\|A_k^+\|_2\|c_k\|_2. \end{aligned} \quad (13.6.21)$$

From its definition, $\bar{d}_k$ is a solution of the subproblem

$$\min_{d \in \mathfrak{R}^n} \quad \bar{g}_k^T d + \frac{1}{2}d^T B_k d, \quad (13.6.22)$$

$$\text{s. t.} \quad A_k^T d = 0, \quad (13.6.23)$$

$$\|\hat{d}_k + d\|_2 \leq \Delta_k. \quad (13.6.24)$$

It is easy to see that $\bar{d}_k$ also solves the calculation

$$\min_{d \in \mathfrak{R}^n} (\bar{P}_k \bar{g}_k)^T d + \frac{1}{2} (\bar{P}_k d)^T B_k (\bar{P}_k d), \quad (13.6.25)$$

$$\text{s. t.} \quad \|d\|_2 \leq \bar{\Delta}_k. \quad (13.6.26)$$

Similar to the proof of Lemma 6.1.3, we can show that

$$\bar{g}_k^T \bar{d}_k \leq -\frac{1}{2} \|\bar{P}_k \bar{g}_k\|_2 \min \left[\bar{\Delta}_k, \frac{\|\bar{P}_k \bar{g}_k\|_2}{2\|B_k\|_2} \right]. \quad (13.6.27)$$

Hence the definitions of $\lambda_k, \bar{d}_k, \bar{g}_k$, the fact that expression (13.6.25) increases monotonically between $d = \bar{d}_k$ and $d = 0$, and the inequalities (13.6.21) and (13.6.27) imply the bound

$$\begin{aligned} (g_k - A_k \lambda_k)^T d_k &+ \frac{1}{2} d_k^T B_k d_k = \left(g_k + \frac{1}{2} B_k d_k \right)^T \bar{d}_k \\ &= \frac{1}{2} [g_k^T \bar{d}_k + \bar{d}_k^T B_k \bar{d}_k + \bar{g}_k^T \bar{d}_k] \leq \frac{1}{2} g_k^T \bar{d}_k \\ &\leq \frac{1}{2} \bar{g}_k^T \bar{d}_k + \frac{1}{2} \|B_k \hat{d}_k\|_2 \|\bar{d}_k\|_2 \\ &\leq -\frac{1}{4} \|\bar{P}_k \bar{g}_k\|_2 \min \left[\bar{\Delta}_k, \frac{\|\bar{P}_k \bar{g}_k\|_2}{2\|B_k\|_2} \right] \\ &\quad + \|A_k^+\|_2 \|B_k\|_2 \|d_k\|_2 \|c_k\|_2. \end{aligned} \quad (13.6.28)$$

Moreover, due to the definition of $\lambda(x)$ and Assumption 13.6.2, there exists a positive constant $\delta_2 > 0$ such that the condition

$$\|\lambda(x_k) - \lambda(x_k + d_k)\|_2 \leq \delta_2 \|d_k\|_2 \quad (13.6.29)$$

holds for all k . The convexity of $\|c_k + A_k^T d\|_2$ shows that

$$\left\| c_k + \frac{1}{2} A_k^T d_k \right\|_2 \leq \frac{1}{2} (\|c_k\|_2 + \|c_k + A_k^T d_k\|_2) \leq \|c_k\|_2. \quad (13.6.30)$$

Therefore, the inequality (13.6.17) now follows from (13.6.7) and (13.6.14) and (13.6.28)-(13.6.30) if we let $\delta_1 = \delta_2 + \sup_{k \geq 1} \{\|B_k\|_2 \|A_k^+\|_2\}$, which is finite due to Assumption 13.6.2. $\square$

A direct corollary of the above lemma is that (13.6.10) is satisfied if $\|c_k\|_2 / \Delta_k$ is sufficiently small.

Corollary 13.6.5 *There exist positive constants δ_3 and δ_4 , such that, on the iterations that satisfy the condition*

$$\|c_k\|_2 \leq \delta_3 \Delta_k, \quad (13.6.31)$$

we have the inequality

$$Pred_k \geq \frac{1}{2} \sigma_k [\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2] + \delta_4 \Delta_k. \quad (13.6.32)$$

Proof. From Assumption 13.6.2, there exists $\bar{M}$ such that $\Delta_k \leq \bar{M}$. If $\delta_3 < \frac{\epsilon}{3\bar{M}}$, (13.6.31) implies $\|c_k\|_2 \leq \epsilon/3$. Unless the algorithm terminates, we have that

$$\|g_k - A_k \lambda_k\|_2 \geq 2\epsilon/3. \quad (13.6.33)$$

If $\delta_3 < \epsilon/(6\bar{M} \sup_k \|B_k\|_2 \|A_k^+\|_2)$, (13.6.31) yields

$$\|c_k\|_2 \leq \frac{\epsilon}{6 \sup_{1 \leq k} \|B_k\|_2 \|A_k^+\|_2}, \quad (13.6.34)$$

which implies that

$$\begin{aligned} \|g_k - A_k \lambda_k\|_2 &= \|\bar{P}_k g_k\|_2 \leq \|\bar{P}_k \bar{g}_k\|_2 + \|\bar{P}_k B_k \hat{d}_k\|_2 \\ &\leq \|\bar{P}_k \bar{g}_k\|_2 + 2\|A_k^+\|_2 \|B_k\|_2 \|c_k\|_2 \\ &\leq \|\bar{P}_k \bar{g}_k\|_2 + \frac{\epsilon}{3}. \end{aligned} \quad (13.6.35)$$

Thus, provided that

$$\delta_3 < \frac{\epsilon}{3\bar{M}} \min \left[1, \frac{1}{2 \sup \|B_k\|_2 \|A_k^+\|_2} \right], \quad (13.6.36)$$

we have, using (13.6.33) and (13.6.35), that

$$\|\bar{P}_k \bar{g}_k\|_2 \geq \frac{\epsilon}{3}. \quad (13.6.37)$$

Consequently, it follows from Lemma 13.6.4 that

$$\begin{aligned} Pred_k &- \frac{1}{2} \sigma_k [\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2] + \delta_1 \|d_k\|_2 \|c_k\|_2 \\ &\geq \frac{\epsilon}{12} \min \left[\bar{\Delta}_k, \frac{\epsilon}{6\|B_k\|} \right]. \end{aligned} \quad (13.6.38)$$

If δ_3 satisfies

$$\delta_3 \leq 0.3 / \sup_k \|A_k^+\|, \quad (13.6.39)$$

we have $\bar{\Delta}_k > 0.8\Delta_k$ from (13.6.31). When

$$\delta_3 < \frac{\epsilon}{24\delta_1} \min \left[\frac{0.8}{\bar{M}}, \frac{\epsilon}{6\bar{M}^2 \sup_k \|B_k\|_2} \right], \quad (13.6.40)$$

it follows from (13.6.31) that

$$\delta_1 \|c_k\|_2 \|d_k\|_2 \leq \frac{\epsilon}{24} \min \left[0.8\Delta_k, \frac{\epsilon}{6\|B_k\|_2} \right]. \quad (13.6.41)$$

Now, inequalities (13.6.38) and (13.6.41) give that

$$\begin{aligned} Pred_k &= \frac{1}{2} \sigma_k [\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2] \\ &\geq \frac{\epsilon}{24} \min \left[0.8\Delta_k, \frac{\epsilon}{6\|B_k\|_2} \right]. \end{aligned} \quad (13.6.42)$$

The corollary follows from the above inequality, and the assumptions that $\{\Delta_k\}$, $\{\|B_k\|\}$ are bounded. $\square$

Now, using the above results, we can easily prove the boundedness of the sequence $\{\sigma_k\}$, which is important in establishing the convergence properties of the algorithm.

Lemma 13.6.6 *The sequence $\{\sigma_k | k = 1, 2, \dots\}$ remains bounded. In other words, because any increase in σ_k is by at least a factor of 2, there exists $\bar{k}$, such that*

$$\sigma_k = \sigma_{\bar{k}}, \quad \forall k \geq \bar{k}. \quad (13.6.43)$$

Proof. Corollary 13.6.5 shows that (13.6.10) fails only if $\|c_k\|_2 > \delta_3 \Delta_k$. In this case, using $\Delta_k \geq \|d_k\|_2$ too, Lemma 13.6.4 provides the bound

$$\begin{aligned} Pred_k &= \frac{1}{2} \sigma_k (\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2) \\ &\geq \|d_k\|_2 \|c_k\|_2 \left[\frac{1}{2} \sigma_k \min(\delta_3, b_2/\delta_5) - \delta_1 \right], \end{aligned} \quad (13.6.44)$$

where δ_5 is an upper bound on $\{\|A_k^+\|_2, k = 1, 2, \dots\}$. Hence condition (13.6.10) holds if $\sigma_k > 2\delta_1 \max[1/\delta_3, \delta_5/b_2]$. Therefore the number of increase in σ_k is finite. $\square$

We now assume without loss of generality that $\sigma_k \equiv \sigma$ for all k . The next lemma shows that both the trust-region bound and the constraints converges to zero, if the algorithm does not terminate after finitely many iterations.

Lemma 13.6.7 *If the algorithm does not terminate, we have the limits*

$$\lim_{k \rightarrow \infty} \Delta_k = 0, \quad (13.6.45)$$

$$\lim_{k \rightarrow \infty} \|c_k\|_2 = 0. \quad (13.6.46)$$

Proof. To prove (13.6.45), we assume that the number

$$\eta = \limsup_{k \rightarrow \infty} \Delta_k \quad (13.6.47)$$

is positive and deduce a contradiction. Define K to be the set of integers k satisfying

$$r_k \geq 0.1, \quad \Delta_k \geq \eta/8. \quad (13.6.48)$$

The set K contains infinitely many elements because of (13.6.47). Since the monotonically decreasing sequence $\{P(x_k)\}$ is convergent, we have that

$$\lim_{\substack{k \in K \\ k \rightarrow \infty}} Pred_k = 0. \quad (13.6.49)$$

Therefore, (13.6.32) does not hold for sufficiently large $k \in K$. It follows from Corollary 13.6.5 and (13.6.48) that

$$\|c_k\|_2 > \delta_3 \eta/8 \quad (13.6.50)$$

holds for all sufficiently large $k \in K$. Thus, Lemma 13.6.3 implies that

$$\begin{aligned} Pred_k &\geq \frac{1}{2} \sigma[\|c_k\|_2^2 - \|c_k + A_k^T d_k\|_2^2] \\ &\geq \frac{1}{2} \sigma\|c_k\|_2 \min \left[\|c_k\|_2, \frac{b_2 \Delta_k}{\|A_k^+\|_2} \right]. \end{aligned} \quad (13.6.51)$$

Using the above inequality, relations (13.6.48) and (13.6.49), we can deduce that

$$\lim_{\substack{k \in K \\ k \rightarrow \infty}} \|c_k\|_2 = 0, \quad (13.6.52)$$

which contradicts (13.6.50). Therefore (13.6.45) is true.

As for (13.6.46), we deduce a contradiction from the assumption that

$$\bar{\eta} = \limsup_{k \rightarrow \infty} \|c_k\|_2 > 0. \quad (13.6.53)$$

Define $\bar{K} = \{k \mid \|c_k\|_2 > \bar{\eta}/2\}$. It follows from (13.6.51) and (13.6.45) that there exists a constant $\bar{\delta} > 0$ such that

$$\text{Pred}_k \geq \bar{\delta} \Delta_k, \quad \forall k \in \bar{K}. \quad (13.6.54)$$

The above inequality and (13.6.45) imply that

$$\lim_{\substack{k \in \bar{K} \\ k \rightarrow \infty}} r_k = 1, \quad (13.6.55)$$

which, together with (13.6.54), shows that

$$\sum_{k \in \bar{K}} \Delta_k < +\infty. \quad (13.6.56)$$

From the definition of $\bar{K}$, inequality (13.6.56) and the continuity of $c(x)$, we can show that

$$\lim_{k \rightarrow \infty} \|c_k\|_2 = \bar{\eta}. \quad (13.6.57)$$

Thus, $k \in \bar{K}$ for all sufficiently large k . This observation and relation (13.6.55) imply that $\Delta_{k+1} \geq \Delta_k$ for all sufficiently large k . This contradicts (13.6.45). Therefore, (13.6.46) is true. $\square$

Having established the above results, we can easily show the global convergence of the algorithm.

Theorem 13.6.8 *Under Assumption 13.6.2, Algorithm 13.6.1 will terminate after finitely many iterations. In other words, if we remove the convergence test from Step 2, then $d_k = 0$ for some k or the limit*

$$\liminf_{k \rightarrow \infty} [\|c_k\|_2 + \|\bar{P}_k g_k\|_2] = 0 \quad (13.6.58)$$

is obtained, which ensures that $\{x_k, k = 1, 2, \dots\}$ is not bounded away from stationary points of the problem (13.4.1)-(13.4.2).

Proof. First we assume that $\epsilon > 0$. If the algorithm does not terminate, then the inequality

$$\|c_k\|_2 + \|\bar{P}_k g_k\|_2 \geq \epsilon \quad (13.6.59)$$

holds for all k . It follows from (13.6.46) that

$$\|\bar{P}_k g_k\|_2 \geq \epsilon/2 \quad (13.6.60)$$

holds for all sufficiently large k . Using (13.6.60), (13.6.45), (13.6.46) and (13.6.17) we can show that there exists a positive constant δ such that

$$Pred_k \geq \delta \Delta_k \quad (13.6.61)$$

is true for all sufficiently large k . The above inequality implies that

$$\lim_{k \rightarrow \infty} r_k = 1, \quad (13.6.62)$$

which leads to the inequality $\Delta_{k+1} \geq \Delta_k$ (for all sufficiently large k). This contradicts (13.6.45). The contradiction indicates that for any positive $\epsilon > 0$ Algorithm 13.6.1 will terminate after finitely many iterations.

If $\epsilon = 0$, then the algorithm terminates if and only if $d_k = 0$. If $d_k = 0$, then x_k is a KKT point of the optimization problem (13.4.1)–(13.4.2). Assume that the algorithm does not terminate, then $d_k \neq 0$ for all k . Let

$$\eta = \inf_k [\|c_k\|_2 + \|\bar{P}_k g_k\|_2]. \quad (13.6.63)$$

If $\eta > 0$, we see that the algorithm does not terminate for $\epsilon = \eta/2$, which contradicts the proof given above. This shows that we must have $\eta = 0$, which implies (13.6.58). $\square$

Under second-order sufficient conditions and other mild conditions, locally superlinear convergence of the algorithm can be proved (see, Powell and Yuan [278]).

Exercises

1. Prove that the trust-region subproblem

$$\min_{d \in \mathbb{R}^n} g_k^T d + \frac{1}{2} d^T B_k d + \sigma_k \| (c_k + A_k^T d)^{(-)} \|_\infty$$

subject to

$$\|d\|_\infty \leq \Delta_k$$

can be reformulated as a quadratic programming problem.

2. Extend the null space trust-region method for equality constrained optimization to handle also inequality constraints.

3. Consider the CDT subproblem when

$$g = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}, \quad A = I, \quad c = \begin{pmatrix} -2 \\ 0 \end{pmatrix},$$

$\Delta = 2$ and $\xi = 1$. Verify that the Hessian of the Lagrange function can have one negative eigenvalue even when only one of the constraints are active at the solution.

4. Construct an example to show that the Hessian of the Lagrange of the CDT subproblem at the solution may have two negative eigenvalues.

5. Let $C, D \in \mathbb{R}^{n \times n}$ be two symmetric matrices and let A and B be two closed sets in $\mathbb{R}^n$ such that $A \cup B = \mathbb{R}^n$. If we have that

$$x^T C x \geq 0, \forall x \in A, \quad x^T D x \geq 0, \forall x \in B,$$

prove that there exists a $t \in [0, 1]$ such that the matrix $tC + (1 - t)D$ is positive semi-definite.

6. Discuss the local convergence properties of Powell-Yuan's trust-region algorithm.